

Data-Driven ILC, Norm-Optimal ILC, and Discrete Repetitive Processes

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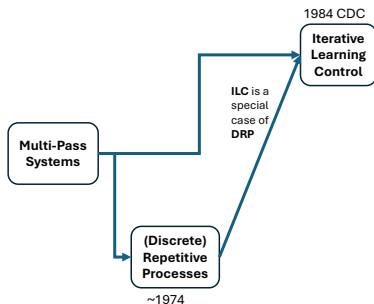
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- **Bing Chu and Paolo Rapisarda** – “Data-driven iterative learning control for continuous-time systems,” in *Proceedings of the 62nd IEEE Conference on Decision and Control*, Singapore, 2023
- **Zheng Jiang and Bing Chu** – “Norm optimal iterative learning control: A data driven approach,” in *Proceedings of 14th IFAC Workshop on Adaptive and Learning Control Systems-ALCOS 2022*, Casablanca, Morocco, 2022
- **Kevin L. Moore, Bing Chu, and Paolo Rapisarda** – “Data-Driven Norm-Optimal Control of Discrete Repetitive Processes,” in *Proceedings of the 2024 IEEE Conference on Decision and Control*, Milan, Italy, 2024

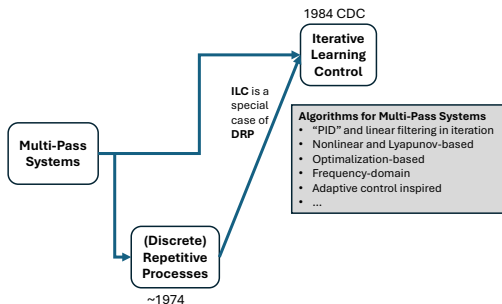
Today's Story



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Philosophically: ILC is a model-free, data-driven paradigm

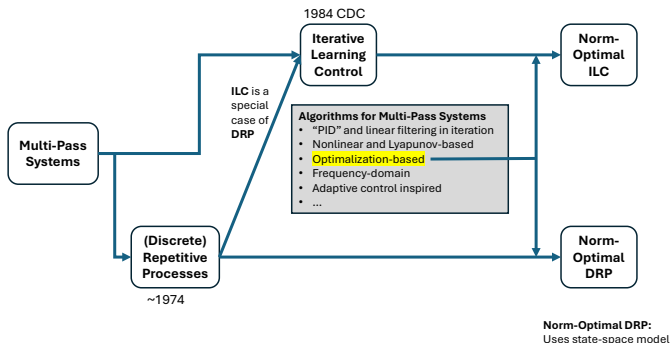
In Practice: Some algorithms need and use models



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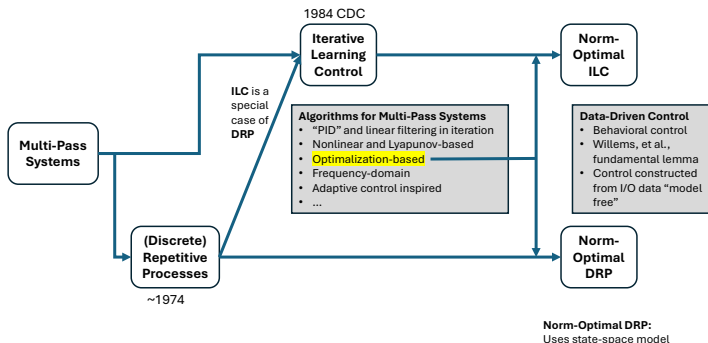
Norm-Optimal ILC:
Uses state-space model



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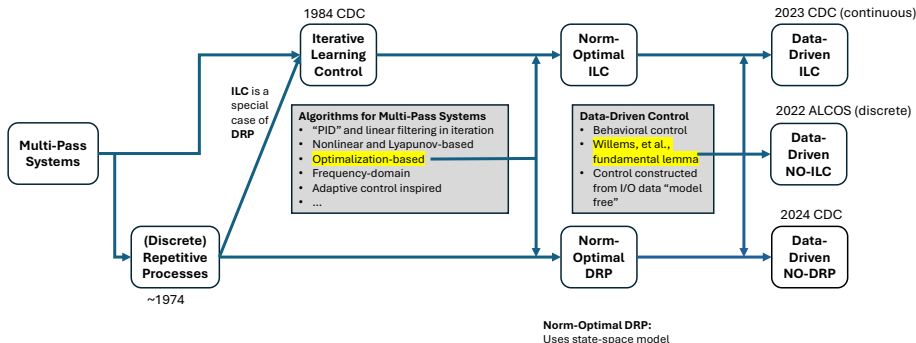


Today's Story

Philosophically: ILC is a model-free, data-driven paradigm
In Practice: Some algorithms need and use models

Norm-Optimal ILC:
 Uses state-space model

Data-Driven NO-ILC/DRP
 Uses only I/O data



Problem Formulation

Consider the following single-input single-output (SISO) discrete time linear time-invariant (LTI) system

$$\begin{aligned}x_k(t+1) &= Ax_k(t) + Bu_k(t), \quad x_k(0) = x_0 \\ y_k(t) &= Cx_k(t)\end{aligned}$$

where k denotes trial index

- This system works repetitively to track a reference sequence $r(t)$
- After t reaches N , a new iteration begins: t will be reset to 0 and the system state will be reset to the same initial condition, i.e., $x_k(0) = x_0$
- The system is required to track the reference $r(t)$ again

Problem Formulation

System Representation: The lifted form system can be represented as

$$Y_k = GU_k + D$$

where the input and output sequence u_k and y_k are defined as

$$U_k = [u_k(0), u_k(1), \dots, u_k(N-1)]^T$$

$$Y_k = [y_k(1), y_k(2), \dots, y_k(N)]^T$$

The system model G ($CB \neq 0$) and the system initial response d can be represented as

$$G = \begin{bmatrix} CB & 0 & 0 & \cdots & 0 \\ CAB & CB & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ CA^{N-1}B & CA^{N-2}B & CA^{N-3}B & \cdots & CB \end{bmatrix}, D = \begin{bmatrix} CAx_0 \\ CA^2x_0 \\ \vdots \\ CA^Nx_0 \end{bmatrix}$$

Problem Formulation

The reference r is defined as

$$R = [r(1), r(2), \dots, r(N)]^T$$

and the tracking error E_k at iteration k is given by

$$E_k = R - Y_k = R - G_k - D$$

Control Objective: To find a control updating law

$$U_{k+1} = f(U_k, E_k)$$

such that

$$\lim_{k \rightarrow \infty} E_k = 0$$

Norm Optimal Iterative Learning Control

The norm optimal iterative learning control (NOILC) algorithm updates the input

$$U_{k+1} = \arg \min_{U_{k+1}} \|E_{k+1}\|_Q^2 + \|U_{k+1} - U_k\|_R^2$$

subject to

$$E_{k+1} = Y - Y_{k+1} = R - GU_{k+1} - D$$

The optimal solution is

$$U_{k+1} = U_k + (G^T Q G + R)^{-1} G^T Q E_k$$

Monotonic Convergence Result

Theorem 1: Convergence properties of NOILC ¹

The NOILC algorithm can achieve monotonic convergence of the tracking error norm to zero

$$\|E_{k+1}\|_Q \leq \|E_k\|_Q, \quad \lim_{k \rightarrow \infty} E_k = 0.$$

¹Amann, N., Owens, D.H., and Rogers, E. (1996). Iterative learning control for discrete-time systems with exponential rate of convergence. IEE Proceedings-Control Theory and Applications, 143(2), 217–224.

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Limitation: NOILC requires the **system model**, i.e. G , in the control updating law. However, the system model G may be difficult/expensive to get in practice.

¹ Amann, N., Owens, D.H., and Rogers, E. (1996). Iterative learning control for discrete-time systems with exponential rate of convergence. IEE Proceedings-Control Theory and Applications, 143(2), 217–224.

Willems' fundamental lemma -1

All T length long trajectories $w := \text{col}(u, y)$ generated by $y = Gu + d$ form a space $\mathcal{G}_T$.

$$\mathcal{G}_T := \left\{ \begin{bmatrix} u \\ y \end{bmatrix} \in \mathbb{R}^{2T} \mid \exists x(t) \in \mathbb{R}^n, \text{ s.t. } \begin{aligned} x(t+1) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) \end{aligned} \right.$$

Hankel matrix $\mathcal{H}_t(x)$ of a signal $x \in \mathbb{R}^T$ is defined as

$$\mathcal{H}_t(x) = \begin{bmatrix} x(0) & x(1) & \cdots & x(T-t) \\ x(1) & x(2) & \cdots & x(T-t+1) \\ \vdots & \vdots & \ddots & \vdots \\ x(t-1) & x(t) & \cdots & x(T-1) \end{bmatrix}$$

The signal x is called **persistently exciting of order t** if $\mathcal{H}_t(x)$ has full row rank

Willems' fundamental lemma -2

Theorem 2: Willems' fundamental lemma ^{1,2}

Consider a controllable system. Given a T sample long data trajectory $w_d := \text{col}(u_d, y_d)$. If u_d is persistently exciting of order $t + n$, then any t sample long system behaviour can be represented for some g as

$$\begin{bmatrix} \mathcal{H}_t(u_d) \\ \mathcal{H}_t(y_d) \end{bmatrix} g = \begin{bmatrix} u \\ y \end{bmatrix}$$

¹Willems, J.C., Rapisarda, P., Markovsky, I., and De Moor, B.L. (2005). A note on persistency of excitation. *Systems & Control Letters*, 54(4), 325–329

²Markovsky, I. and Rapisarda, P. (2008). Data-driven simulation and control. *International Journal of Control*, 81(12), 1946–1959.

Willems' fundamental lemma -3

Data-driven simulation using the fundamental lemma: given initial trajectory (u^{ini}, y^{ini}) , the output response y to a known input u can be computed by solving

$$\begin{bmatrix} U_p \\ Y_p \\ U_f \\ Y_f \end{bmatrix} g = \begin{bmatrix} u^{ini} \\ y^{ini} \\ u \\ y \end{bmatrix}$$

where U_p , Y_p , U_f and Y_f are defined by existing data as

$$U = \mathcal{H}_t(u) = \begin{bmatrix} U_p \\ \vdots \\ U_f \end{bmatrix} \quad Y = \mathcal{H}_t(y) = \begin{bmatrix} Y_p \\ \vdots \\ Y_f \end{bmatrix}$$

Data-driven NOILC

Given a existing data w_d and assume the input u_d is persistently exciting of order $N + 2n$. The input u_{k+1} can be calculated by

$$u_{k+1} = \arg \min_{u_{k+1}} \|e_{k+1}\|_Q^2 + \|u_{k+1} - u_k\|_R^2$$

$$\text{s.t. } \begin{bmatrix} U_p \\ Y_p \\ U_f \\ Y_f \end{bmatrix} g = \begin{bmatrix} \mathbf{0}_{n,1} \\ \mathbf{0}_{n,1} \\ u_{k+1} - u_k \\ y_{k+1} - y_k \end{bmatrix}, \quad e_{k+1} = r - y_{k+1}.$$

Thus the NOILC design problem can be solved without system model G .

Data-driven NOILC

The data-driven NOILC solution is given by

$$u_{k+1} = u_k + \begin{bmatrix} I \\ \mathbf{0}_{N,N} \end{bmatrix}^T W_0 (W_0^T S W_0)^\dagger W_0^T S \begin{bmatrix} \mathbf{0}_{N,1} \\ e_k \end{bmatrix},$$

where $\dagger$ denotes the pseudo-inverse, $S = \text{diag}\{Q, R\}$ and W_0 is a matrix constructed from data.

Theorem 3: Convergence properties of Data-driven NOILC

The tracking performance of data-driven NOILC is identical to the model-based NOILC. Consequently, the proposed algorithm can guarantee

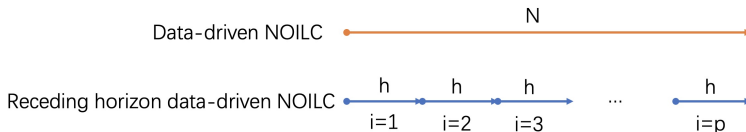
$$\|e_{k+1}\|_Q \leq \|e_k\|_Q, \quad \lim_{k \rightarrow \infty} e_k = 0.$$

Receding Horizon Based Data-driven NOILC -1

- **Limitation:** When the trial length is long the persistently exciting assumption (of order $N + 2n$) may be hard to achieve

Receding Horizon Based Data-driven NOILC -1

- **Limitation:** When the trial length is long the persistently exciting assumption (of order $N + 2n$) may be hard to achieve
- One solution is to use a receding horizon approach



$$\begin{aligned}
 u_{k+1} &= \arg \min_{u_{k+1}} \sum_{t=0}^{N-1} \|e_{k+1}(t+1)\|_Q^2 + \|u_{k+1}(t) - u_k(t)\|_R^2 \\
 &= \arg \min_{u_{k+1}} \sum_{i=0}^p \sum_{t=ih}^{ih+h-1} \|e_{k+1}(t+1)\|_Q^2 + \|u_{k+1}(t) - u_k(t)\|_R^2 \\
 &= \arg \min_{u_{k+1}} \sum_{i=0}^p \|e_{k+1,ih}\|_Q^2 + \|u_{k+1,ih} - u_{k,ih}\|_R^2
 \end{aligned}$$

Receding Horizon Based Data-driven NOILC -2

The input $u_{k+1,t}$ can be calculated by minimizing the following cost function

$$u_{k+1,t} = \arg \min_{u_{k+1,t}} \|e_{k+1,t}\|_Q^2 + \|u_{k+1,t} - u_{k,t}\|_R^2$$

$$\text{s.t.} \quad \begin{bmatrix} U_p \\ Y_p \\ U_f \\ Y_f \end{bmatrix} g = \begin{bmatrix} u_{k+1,t}^{ini} \\ y_{k+1,t}^{ini} \\ u_{k+1,t} \\ y_{k+1,t} \end{bmatrix}$$

$$e_{k+1,t} = r_t - y_{k+1,t}$$

- This problem can be solved without a system model
- u_d needs to be persistently exciting of order $h + 2n$

Receding Horizon Based Data-driven NOILC -3

The data-driven receding horizon based NOILC solution is given by

$$u_{k+1,t} = u_{k,t} + \begin{bmatrix} I \\ \mathbf{0}_{h,h} \end{bmatrix}^T W_{0h} (W_{0h}^T S W_{0h})^\dagger W_{0h}^T S \begin{bmatrix} \mathbf{0}_{h,1} \\ e_{k,t} + d_{k,t} - d_{k+1,t} \end{bmatrix},$$

where $S = \text{diag}\{Q, R\}$, W_{0h} , $d_{k+1,t}$ are constructed from the data.

Theorem 4: Convergence properties of receding horizon based data-driven NOILC

The proposed algorithm can guarantee the tracking error norm asymptotically converges to zero, i.e. $\lim_{k \rightarrow \infty} e_k = 0$. Moreover, monotonic convergence ($\|e_{k+1}\|_Q \leq \|e_k\|_Q$, $\lim_{k \rightarrow \infty} e_k = 0$) can be achieved for scalar weightings $Q = q \times I$, $R = r \times I$ when the ratio q/r is sufficiently large.

Receding Horizon Based Data-driven NOILC -4

Table: Comparison between different algorithms

	Receding-horizon data-driven NOILC	Data-driven NOILC	Model-based NOILC
System model	Not required	Not required	Required
Persistently exciting order	$h + 2n$	$N + 2n$	-
Convergence properties	Asymptotic /Monotonic	Monotonic	Monotonic

Simulation Examples

Consider a gantry robot z-axis whose transfer function is

$$H(s) = \frac{15.8869(s + 850.3)}{s(s^2 + 707.6s + 3.377 \times 10^5)}.$$

The system is sampled with a sampling time of $T_s = 0.01$ s. The trial length $N = 200$. We simulate the system to track the reference signal

$$r = 0.005 \sin(2\pi t - \frac{\pi}{2}) + 0.005 \quad t \in [0, 2].$$

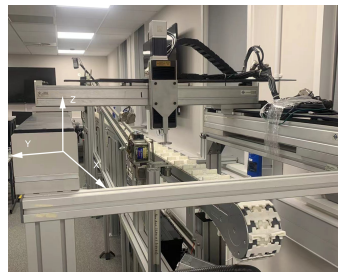


Figure: Three-axis gantry robot

Simulation Examples

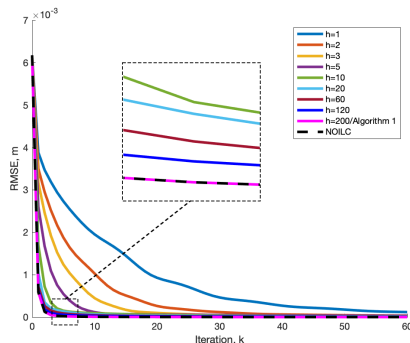


Figure: Root mean square error for different horizon h

Simulation Examples

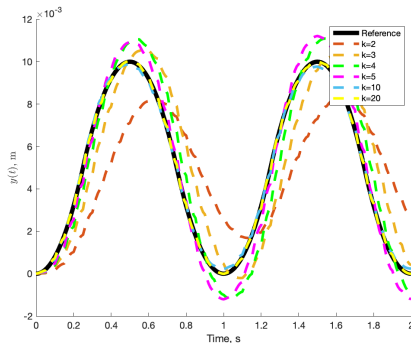


Figure: Reference and output for different trials of receding horizon solution when $h = 5$

Simulation Examples

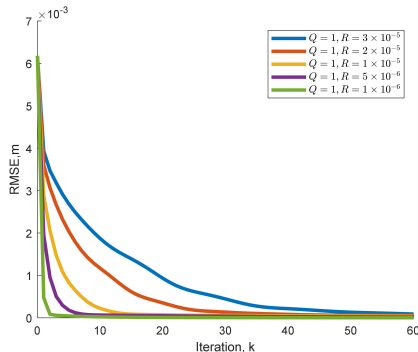


Figure: Root mean square error for different Q and R of receding horizon solution when $h = 5$

DRP Plant

- Consider the DRP plant

$$\begin{aligned}x_{k+1}(t+1) &= Ax_{k+1}(t) + Bu_{k+1}(t) + B_0y_k(t) \\y_{k+1}(t) &= Cx_{k+1}(t) + Du_{k+1}(t) + D_0y_k(t)\end{aligned}\quad (1)$$

where we assume $x_k(0) = x_0$ for all k

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where we assume $x_k(0) = x_0$ for all k

- Define the lifted *supervectors*

$$\begin{aligned}U_k &:= [u_k(0), u_k(1), \dots, u_k(N-1)]^T \\ Y_k &:= [y_k(0), y_k(1), \dots, y_k(N-1)]^T\end{aligned}$$

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- Then for $0 \leq t \leq N-1$ and $k > 0$, the 1-D difference equation model in the lifted domain is given by (assuming D, D_0 are not zero)

$$Y_{k+1} = H^u U_{k+1} + H^y Y_k + H^0 x_0$$

Lifted Plant Matrices

- H^u and H^y are the Toeplitz matrices (of rank N since $D, D_0 \neq 0$)

$$H^u := \begin{bmatrix} D & 0 & \cdots & 0 \\ CB & D & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ CA^{N-1}B & \cdots & CB & D \end{bmatrix} \quad H^y := \begin{bmatrix} D_0 & 0 & \cdots & 0 \\ CB_0 & D_0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ CA^{N-1}B_0 & \cdots & CB_0 & D_0 \end{bmatrix}$$

- H^0 is defined by:

$$H^0 := \begin{bmatrix} C^T & (CA)^T & \cdots & (CA^{N-1})^T \end{bmatrix}^T$$

Norm-Optimal DRP

Algorithm 1: On iteration $k + 1$, the following optimization problem is solved to obtain the input supervector U_{k+1} :

$$U_{k+1} = \arg \min_{U_{k+1}} \{ \|E_{k+1}\|_Q^2 + \|U_{k+1} - U_k\|_R^2 \},$$

subject to

$$E_{k+1} = r - Y_{k+1}$$

$$Y_{k+1} = H^u U_{k+1} + H^y Y_k + H^0 x_0$$

where the associated norms are defined by

$$\|E_{k+1}\|_Q^2 := E_{k+1}^T Q E_{k+1}$$

$$\|U_{k+1} - U_k\|_R^2 := (U_{k+1} - U_k)^T R (U_{k+1} - U_k)$$

Solution of Algorithm 1

- The solution to Algorithm 1 can be shown to be (using two previous iterations):

$$U_{k+1} = U_k + (H^u{}^T Q H^u + R)^{-1} \\ \times H^u{}^T Q (E_k - H^y(Y_k - Y_{k-1}))$$

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- This compares nicely to the ILC case. where $H^y=0$:

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$$U_{k+1} = U_k + (H^u{}^T Q H^u + R)^{-1} H^u{}^T Q E_k$$

- The NO-DRP solution can also be equivalently written as (using initial conditions and one previous iteration):

$$U_{k+1} = U_k + (H^u{}^T Q H^u + R)^{-1} H^u{}^T Q [E_k \\ + (I - H^y) Y_k - H^u U_k - H^0 x_0]$$

Properties of Algorithm 1

- Note that $E_{k+1} = L_1 E_k + L_2 E_{k-1}$, where the learning operators L_1, L_2 are defined by

$$L_1 := (H^u R^{-1} H^{u^T} Q + I)^{-1} (I + H^y)$$

$$L_2 := -(H^u R^{-1} H^{u^T} Q + I)^{-1} H^y$$

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- Theorem 1:* Algorithm 1 has the following properties:
 - ① $\lim_{k \rightarrow \infty} E_k = 0$ for every initial tracking error E_0 iff

$$\rho \left(\begin{bmatrix} L_1 & L_2 \\ I & 0 \end{bmatrix} \right) < 1$$

- ② Under this condition, the input U_k converges to

$$\lim_{k \rightarrow \infty} U_k = U_\infty = H^{u-1} ((I - H^y)r - H^0 x_0)$$

State-Space Fundamental Lemma

From [1] I. Markovsky and P. Rapisarda, “Data-driven simulation and control,” *International Journal of Control*, vol. 81, no. 12, pp. 1946–1959, 2008:

- Consider the system

$$\begin{aligned}x(t+1) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}\tag{2}$$

- Denote by w_T the T samples-long i - o system trajectory

$$\begin{bmatrix} u(0)^T & \dots & u(T-1)^T & y(0)^T & \dots & y(T-1)^T \end{bmatrix}^T$$

State-Space Fundamental Lemma (cont.)

- Define the *Hankel matrix of depth t* of a signal f_T (for sample length T large enough) by

$$\mathcal{H}_t(f_T) = \begin{bmatrix} f(0) & f(1) & \cdots & f(T-t) \\ f(1) & f(2) & \cdots & f(T-t+1) \\ \vdots & \vdots & \ddots & \vdots \\ f(t-1) & f(t) & \cdots & f(T-1) \end{bmatrix}$$

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- A signal f_T is called *persistently exciting of order t* if $\mathcal{H}_t(f_T)$ has full row rank

State-Space Fundamental Lemma (cont.)

The *Fundamental Lemma* introduced in the behavioral setting can be expressed in the state space as follows:

Theorem 2 [Lemma 2 of [1]]: Assume that the system (2) is controllable and let $w_{d,T} = \begin{pmatrix} u_{d,T} \\ y_{d,T} \end{pmatrix}$ be a T -sample length i-o trajectory generated by (2). If $u_{d,T}$ is persistently exciting of order $t + n$, where n is the order of (2), then

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- ① Every linear combination of the columns of $\mathcal{H}_t(w_{d,T})$ is a t sample length of a system trajectory
- ② For every t sample length trajectory $w_t = \begin{pmatrix} u_t \\ y_t \end{pmatrix}$ there exists $g \in \mathbb{R}^{T-t+1}$ such that $\mathcal{H}_t(w_{d,T})g = w_t$

Fundamental Lemma for DRPs

We can apply the Fundamental Lemma (Theorem 2 above) to develop an equivalent result for discrete-time repetitive processes:

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Theorem 3: Consider the unit memory discrete-time repetitive process (1) and assume that the pair $(A, [B \ B_0])$ is controllable. Given a T sample length trajectory

$$w_{d,T} = [u_{d,T}^T, (y_{d,T}^-)^T, y_{d,T}^T]^T$$

of (1), where $u_{d,T}, y_{d,T}^-, y_{d,T}$ are input, past iteration profile data, and output,

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Theorem 3: Consider the unit memory discrete-time repetitive process (1) and assume that the pair $(A, [B \ B_0])$ is controllable. Given a T sample length trajectory

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of (1), where $u_{d,T}, y_{d,T}^-, y_{d,T}$ are input, past iteration profile data, and output, if $\begin{pmatrix} u_{d,T} \\ y_{d,T}^- \end{pmatrix}$ is persistently exciting of order $t+n$, then on iteration $k+1$ for every past iteration profile y_k , for any t sample length trajectory $w_t = \begin{pmatrix} u_{t,k+1} \\ y_{t,k+1} \end{pmatrix}$ of (1) there exists $g \in \mathbb{R}^{T-t+1}$ such that

$$\begin{bmatrix} u_{t,k+1} \\ y_{t,k} \\ y_{t,k+1} \end{bmatrix} = \begin{bmatrix} \mathcal{H}_t(u_{d,T}) \\ \mathcal{H}_t(y_{d,T}^-) \\ \mathcal{H}_t(y_{d,T}) \end{bmatrix} g$$

Data-Driven NO DRP

Theorem 4: Assume that the pair $(A, [B \ B_0])$ in the unit-memory discrete-time repetitive process (1) is controllable. Suppose we have a T sample length system trajectory $w_{d,T} = [u_{d,T}^T, (y_{d,T}^-)^T, y_{d,T}^T]^T$ of (1).

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$$U_{k+1} = \arg \min_{U_{k+1}} \{ \|E_{k+1}\|_Q^2 + \|U_{k+1} - U_k\|_R^2 \},$$

subject to

$$E_{k+1} = r - Y_{k+1}$$

$$\begin{bmatrix} 0_n \\ U_{k+1} - U_k \\ 0_n \\ Y_k - Y_{k-1} \\ 0_n \\ Y_{k+1} - Y_k \end{bmatrix} = \begin{bmatrix} \mathcal{H}_{N+n}(u_{d,T}) \\ \mathcal{H}_{N+n}(y_{d,T}^-) \\ \mathcal{H}_{N+n}(y_{d,T}) \end{bmatrix} g$$

for some $g \in \mathbb{R}^{T-t+1}$ produces the same U_{k+1} as Algorithm 1

Comments

- As in the model-based-case, it is possible to devise a data-driven NO-DRP that depends only on the most recent iteration and initial conditions, rather than on the two most recent iterations
- It is possible to relax the persistency of excitation condition from $N + 2n$ to $q + 2n$, where q is an integer factor of N (a somewhat technical result)
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- This data-driven, norm optimal, DRP algorithm inherits all the convergence properties of the model-based algorithm.
- We stress that the key contribution is that Theorem 4 provides a way to compute the norm-optimal input to the DRP system's next iteration in a **completely model-free fashion, using no knowledge of the plant matrices.**

Implementation

Algorithm 2: The data-driven NO-DRP algorithm has the following steps:

- ① Data Collection: with initial profile Y_d^- , apply an input U_d (e.g., a random signal), store the output Y_d . Construct w_d and check the persistency of excitation condition Theorem 4. Repeat this process with updated pass profile $Y_d^- = Y_d$ until the persistency of excitation condition is met
- ② Set $k = 1$, with initial profile Y_0 , applying an input U_1 and store the system output Y_1
- ③ Solve for U_{k+1} as in Theorem 4
- ④ Apply the input U_{k+1} and store the output Y_{k+1}
- ⑤ If $\|r - Y_{k+1}\| \leq \epsilon$ where ϵ is a prescribed tracking accuracy level, then stop; else, go to Step 3

Example

- Consider the following plant

$$x_{k+1}(t+1) = \begin{bmatrix} 0 & 0.1 \\ 0.125 & -0.2 \end{bmatrix} x_{k+1}(t) + \begin{bmatrix} 0.8 \\ 1 \end{bmatrix} u_{k+1}(t) + \begin{bmatrix} 0.8 \\ 0.5 \end{bmatrix} y_k(t)$$
$$y_{k+1}(t) = [0 \quad 1] x_{k+1}(t) + 1.2u_{k+1}(t) + 0.8y_k(t)$$

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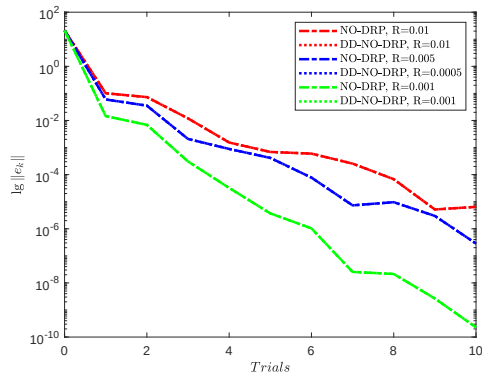
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- We compare NO-DRP with DD-NO-DRP for several values of Q and R

Results

- DD-NO-DRP produces the same results as NO-DRP (their error plots are indistinguishable)
- Weighting matrices have a similar effect on the convergence speed as in NO-ILC; namely, “smaller” R s (keeping Q the same) lead to faster error convergence
- Final converged inputs for all the weighting choices have also been checked and confirmed to be the theoretical solution



Summary-1

- By combining results from several trains of thought we have shown
 - Data driven norm-optimal ILC using the Willems fundamental lemma, along with a computational feasible receding horizon approach
 - A new NO-DRP algorithm (Algorithm 1, Theorem 1)
 - A fundamental lemma for DRP systems (Theorem 3)
 - A data-driven approach to NO control of DRP systems (Theorem 4, Algorithm 2)

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- Not shown is work from CDC 2023 for data-driven, continuous-time ILC using Chebyshev bases
- These model-free data-driven ideas are one approach to achieving the philosophical intent of ILC

Summary -2

Philosophically: ILC is a model-free, data-driven paradigm
In Practice: Some algorithms need and use models

Norm-Optimal ILC:
 Uses state-space model

Data-Driven NO-ILC/DRP
 Uses only I/O data

